

Mid Semester Test-1UID No: 24me10050**Academic Year 2024-2025**

Branch : MCA

Subject Code : 24CAT-671

Subject Title : Machine Learning

Semester : 2

Duration : 1 Hour

Maximum Marks : 20**Instructions : Attempt all questions**

Sl. No.	Statement	CO Mapping	BT Level
Section A 5x2=10 marks			
1	Could you state the key differences between machine learning and traditional programming?	CO1	1
2	Describe the main goal of machine learning. ✓	CO1	2
3	Interpret the slope coefficient in linear regression.	CO2	1
4	Differentiate between regression and classification.	CO2	2
5	List the importance of Machine Learning Algorithms. ✓	CO1	1
Section B 2x5=10 marks			
6	Differentiate error surface learning and explain its significance in machine learning optimization.	CO1	4
7	Differentiate between the various learning techniques used in unsupervised machine learning.	CO3	4

Machine Learning

Accuracy depends on amount of data, as huge amount of data helps build better model which predict accurately

↳ Branch of AI enables computers to learn from data and make predictions or decisions without being explicitly programmed

- Learns pattern from data
- make predictions or decisions
- Improve with experience

Traditional

vs

ML

Rules + Data → output
fixed logic
manual rule creation
Not adaptive

Data + Output → model
Learns Logic from data
Automatic learning
Improves with experience

Why → Automates complex tasks Improves accuracy and efficiency

→ Handles large complex datasets

Key components

1. Data → Foundation of ML models

can be structured or unstructured

Better data → Better prediction

Ex image, Text, Numbers, Videos.

2. model → mathematical structure that learns pattern from data

Ex Linear Regression, Decision Trees, Neural Networks

3. Algorithm $\Rightarrow$ method used to train the model by adjusting parameters to reduce error
 Ex Gradient Descent, Backpropagation, K-means algorithm

Types 1. Supervised Learning process of providing input data as well as correct output data

Adv $\Rightarrow$ ~~comprehension~~ $\rightarrow$ uses labeled data (input + output)
 the basis of prior experiences $\rightarrow$ Good prediction
 $\rightarrow$ exact idea about classes of objects $\rightarrow$ Model learns from known
~~data~~ $\rightarrow$ Not suitable for complex tasks answers.

Techniques Classification, Regression $\rightarrow$ Linear/Non
 cannot predict ~~data~~ correct output $\downarrow$ Logistic Regression
 text data is different from ~~image~~ data

Ex Spam detection, sales prediction, Disease diagnosis

2. Unsupervised Learning

$\rightarrow$ unlabeled data
 $\rightarrow$ Goal - To find hidden patterns
 $\rightarrow$ no predefined output

Techniques - Clustering, Anomaly Detection

Ex - Fraud detection, market analysis

$\rightarrow$ useful for finding useful insights from the data

$\rightarrow$ works on unlabeled and uncategorized data

Adv $\rightarrow$ more complex data Disadv
 $\rightarrow$ more difficult

3. Semisupervised - uses small labeled + large ^{unlabeled} data.

Combines supervised + unsupervised L

→ Useful when labelling data is expensive

Ex image ~~detection~~ recognition with limited labeled image.

4. Reinforcement Learning → Learning through trial and error

→ Uses rewards and penalty

→ Agent interacts with environment

Ex Robotics, Gaming (Chess), Self driving cars

Python libraries for ML

1. Numpy used for numerical computations array and matrix

2. Pandas Data manipulation and analysis
Handles datasets using Data Frames

Data Cleaning and preprocessing

3. Scikit Learn ML library contains algorithms for
built-in many Classification, Regression, clustering
scipy Easy to use used for data mining and data analysis

4. Tensorflow used for Deep learning supports
opensource library neural networks developed by google.

opensource library

Tensorflow, Keras, PyTorch, Pandas, Matplotlib, Scipy
↓
evaluate things and array, mathematical expression
high level neural network applicable of running on top of tensorflow
implemented in C with wrapper in Lua support computation on GPU with
Tensor ~~flow~~ computation on GPU with
linear algebra integration, statistics

Application of machine learning

- Health care → Disease prediction, medical image analysis, Personalized treatment
- Finance → Fraud detection, Credit scoring, stock prediction
- E commerce → Product recommendations, customer segmentation.
- Social media → Face recognition, content filtering, Friend suggestions.
- Transportation → Self driving vehicles, Route optimization
- Education → Personalized learning, Performance prediction, plagiarism detection.
- Cybersecurity → Intrusion detection, Malware classification
- Agriculture → Crop disease detection, Smart irrigation, Yield prediction

Supervised Learning Steps

- Determine the type of training dataset
- Collect labelled training data
- Split the training dataset into training dataset, test dataset and validation dataset
- Determine input features of training dataset
- Determine suitable algo for model
- Execute algo on training dataset
- Evaluate the accuracy of model by providing the test set. If model predicts the correct output which means our model is accurate.

Types

Regression → used if there is relationship b/w input variable and output variable
used for weather forecasting, market-trend etc

EX Linear, non-linear, Regression trees,
Polynomial Regression.

Linear Regression → method used to find relationship b/w two variables and predict values using a straight line

→ finds best ~~line~~ fit line that shows how one variable changes when another changes.

→ estimates how a dependent variable (output) changes based on an independent variable (input) and use relationship to make predictions.

$$Y = mX + C$$

↓ ↓ ↘ intercept (starting point)
 Output Input

(Dependent variable)

Multiple Linear Regression → more than one input

- 1 → Take dataset Ex Hours → marks
- 2 → Draw Random Line ↓ squared difference b/w actual and predicted values
- 3 → Measures Error • MSE • RMSE
- 4 → Adjust line → Using Gradient descent
- 5 → Best Fit line found

improves accuracy by reducing error

Optimization algo
used to minimize (loss error)

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Gradient Descent → methods used in ML to reduce error by slowly adjusting model parameters (like weights) until predictions become accurate

→ works by moving toward the lowest point of a function

used in linear regression, neural networks

- Start with random parameter value
- Calculate error (loss function)
- Calculate gradient (slope of error curve)
- Move parameter opposite to gradient
- Repeat until error is minimum

(*)
$$\text{new parameter} = \text{old parameter} - \text{learning rate} \times \text{gradient}$$

Real life Applications

Adv

- salary prediction
 - Risk analysis
 - Demand prediction
- Easy to understand
 - Fast to train
 - Works well for linear data

disadv

- only works well for linear relationships
- cannot capture complex patterns

Ex	Study hour (X)	Marks (Y)
	1	50
	2	55
	3	65
	4	70
	5	80

Step 1 Observe pattern

As hour increases $\rightarrow$ marks increase
 straight line $\rightarrow$ linear regression

Step 2 Find Line Equation

$Y = mX + C$ Suppose model may

$$Y = 7X + 43$$

Step 3 Prediction

$$Y = 7(6) + 43$$

$$= 42 + 43$$

$$= 85 \text{ marks}$$

Step 4 error checking

Actual marks maybe = 83

$$\text{predicted} = 85$$

$$\text{error} = 2$$

Non Linear Regression

where relationship b/w input and output is not a straight line can be curved.

$$y = f(x, \beta) + \epsilon$$

output
input
model parameters
epsilon

error term

Types

Polynomial Regression - curve using powers of x
 x^2, x^3

Exponential Regression - Used in population growth
 virus spread

Logistic Regression → Used in Disease prediction
 classification but
 curve boundary

Works

- choose curve model type
- estimate parameters
- Use of iterative optimization (like gradient methods)
- minimize error.

Application

- Finance → market trends
- Biology → population growth

Adv

- captures real world complexity

→ Better accuracy for non linear data

Limitations - Hard to interpret

- Needs more computation

- Risk of overfitting

Ex X: 1 2 3 4 5

Y: 5, 20, 45, 80, 125

Curve non linear

Step 1 → Start random equation

Step 2 → Predict values

Step 3 → Compare with actual

Step 4 → Calculate Error

5 → Adjust Equation

6 → Repeat until error minimum.

Linear → High Bias, Low Variance

Non-linear → Low Bias, High Variance

Training error High
Test error High
Error due to
simple model assumptions

→ of model too simple ~~model~~
it cannot learn real patterns →
give wrong predictions

→ of real data is curved model try
straight line → high bias

→ High bias → underfitting
model too simple

Low
high error due to model
being too sensitive to
training data

→ High Variance
overfitting

→ Model too complex

ML Pipeline for Regression

- Collect data
- Clean data
- Split Train/Test
- Train Model
- Evaluate model
- Tune model
- Deploy Model

Classification :→ used when the output variable is categorical

Yes/No, True/False

- Spam Filtering
- Random Forest
- Decision trees
- Logistic Regression

Logistic Regression supervised machine learning algo used for Classification problem

It predicts Yes/No, 0/1, True/False

- output can be < 0 or > 1

But probability $0 \leq \text{probability} \leq 1$

It uses sigmoid function

$$p = \frac{1}{1 + e^{-z}}$$

$$\text{where } z = w_1 x_1 + w_2 x_2 + b$$

range 0 to 1

How it works

- 1 → Take input data ex hours studied → Pass/Fail
- 2 → Apply linear Equation
 $z = mx + c$
- 3 → Apply sigmoid function
convert output into probability
- 4 → Apply Threshold (usually 0.5)
If Probability > 0.5 → Class 1
If Probability < 0.5 → Class 0

Study hours

Pass (1) / Fail (0)

1
2
3
4
5

0
0
0
1
1

Types

1. Binary Logistic Regression
Two classes (Yes/No)
2. Multinomial Logistic Regression
more than two classes. (Red/Yellow/Blue)
without any order
3. Ordinal Logistic Regression
ordered categories (Low/medium/high)
classes have natural order

Adv

- simple
- fast
- works well for binary classification

disadvantages

- Not good for complex patterns

Decision Tree → supervised learning algo
splits data based on conditions
form a tree like structure

used for classification & regression.

Ex of weather = Rain → stay home
else → Go outside

Tree like decision making

How it works

1. Take data set
2. Choose Best Feature
 Using Gini Index
 - Entropy - measures impurity
 - Information Gain
3. Split Data
4. Repeat until Leaf Node

Measure
 node purity
 Splitting
 Criteria

- Root Node
 → Starting decision
- Internal Node
 → Decision condition
- Leaf Node
 → Final output

Entropy reduction
 after split

Adv

disadv

- Easy to understand
- Works with categorical data
- No data normalization needed
- Overfitting possible
- Unstable sometimes

Types of Trees

- Classification Tree
 Output is Categorical
 Ex: Yes/No
- Regression Tree
 Output is Continuous
 Ex: House price

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Confusion matrix shows how your model's predictions compare with actual values.

Binary classification case

	Predicted YES	Predicted NO
Actual YES	TP	FN
Actual NO	FP	TN

True Positive (TP)

Model Predicts YES
Actual result is YES

Ex. • Model says: Disease
• Patient actually has Disease

False Positive (FP)

Model Predicts YES
Actual result is NO

Ex. • Model says disease
• Patient is actually healthy
False alarm.

False Negative (FN)

Model Predicts NO
Actual result is YES

Ex. Model says NO Disease
Patient actually has Disease.

True Negative (TN)

Model Predicts NO
Actual result is NO

Ex. Model says NO
Patient is actually healthy

Logistic

- Classification
- Model nature Mathematical
- Decision Boundary Linear
- Overfitting Less
- Speed Fast

Decision Tree

Classification Regression

Tree Structure

Nonlinear

More

medium

universal approximators - model (usually of
neural networks) that can approximate
any mathematical function of it has
↳ close to exact value

- enough neurons
- enough data
- proper training.

→ It can learn almost any pattern or relationship b/w input and output

Theorem - At least one hidden layer & enough neurons, Non linear proper training, activation function (sigmoid Tanh)
can approximate any continuous function

↳ it means it can learn and copy any smooth pattern b/w input and output if it is big enough and trained well

Application weather prediction

- speech recognition
- image recognition
- medical diagnosis

iterative, recursive and efficient method
set of methods to train artificial neural networks.

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Backpropagation $\Rightarrow$

Derivatives of activation function needed for this
neural networks to update weights by calculating how much each weight contribute to final error

- $\rightarrow$ network makes prediction
- $\rightarrow$ compare with actual answer
- $\rightarrow$ sends error backward
- $\rightarrow$ adjust weights to improve next prediction

$\otimes$ works together with gradient descent to minimize loss

Features to update weights

$\hookrightarrow$ gradient descent method used in case of

simple perceptron network with the differentiable unit $\rightarrow$ single neuron

$\hookrightarrow$ has derivative

- Three stages
 - Feed forward of input training
 - the calculation and backprop of error
 - updating of weight.

Steps

1. Input x arrive through the pre connected

MSE $\rightarrow$ (Predicted output - Actual output)²

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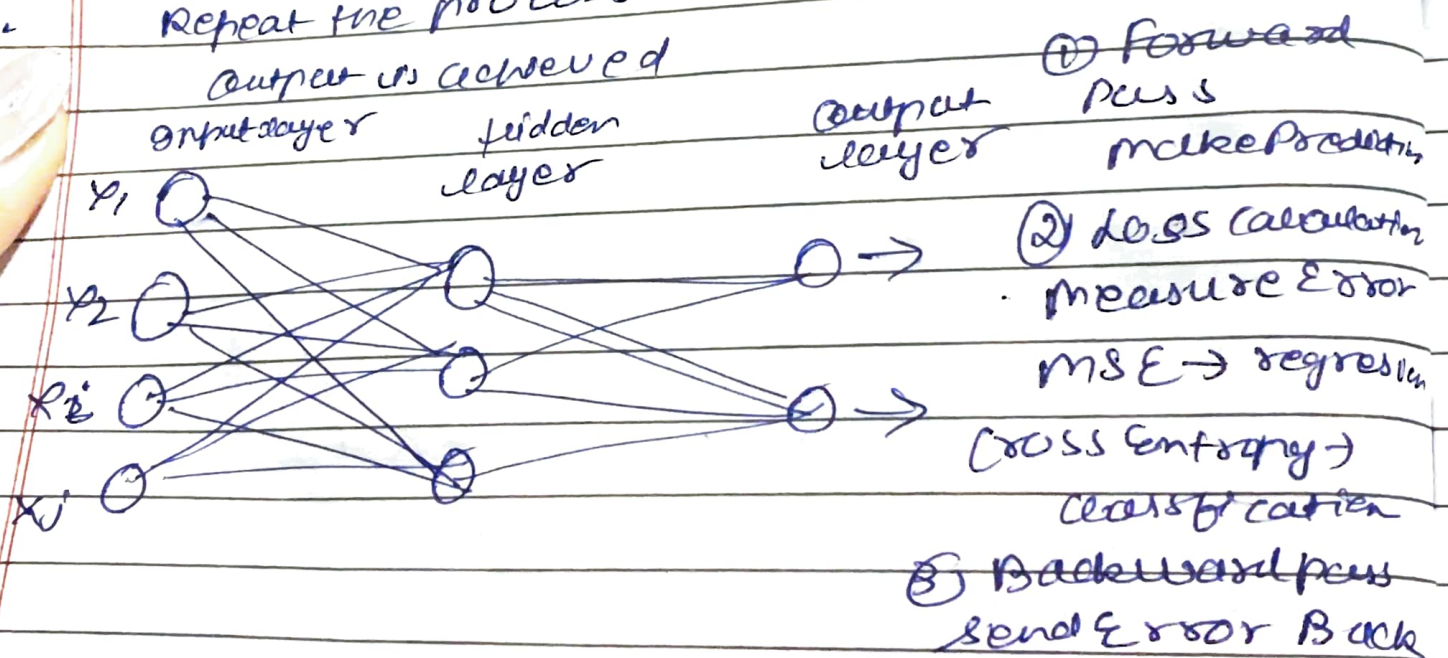
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2. Input in model using true weight w usually chosen randomly
3. Calculate the output of each neuron from input layer to hidden layer to output layer
4. Calculate error in outputs

Backpropagation
 error = Actual output - Desired output

5. From output layer go back to hidden layers to adjust the weights to reduce the error

6. Repeat the process until desired output is achieved



④ weight update

$$\text{New } w = \text{Old } w - \text{Learning Rate} \times \text{Gradient}$$

Why make neural network learn
 work for deep learning models
 Efficient for large networks

Adv

- simple fast easy to program
- flexible efficient

Disadv

- sensitive to noisy data
- performance highly dependent on input data

Error surface learning

- ⇒ The relationship b/w the model parameters and error value can be represented graphically which is known as error surface
- ⇒ process of learning model parameters by navigating this surface to reach the minimum error is error surface learning

Cost functions

MSE → used in regression

Log Loss → used in classification

Steps

- initialize model parameters with random values
- compute predictions using current parameters
- calculate error using the cost function
- move on the error surface towards the minimum

- Repeat until minimum error is achieved
aim to reach global minimum of error surface

Clustering type of unsupervised ML

↳ used to group similar data points together

Data points in same cluster → similar
different → different

How it works

- Take datasets
- measure similarity
- Group closest point together
- form clusters

Ex

- Group customers based on shopping habits
- group students based on marks

<u>ex</u>	<u>Student</u>	<u>Marks</u>
	A	95
	B	90
	C	85
	D	40
	E	35
	F	30

Cluster 1 high scores $\rightarrow (95, 90, 85)$

Cluster 2 $\rightarrow$ low scores $\rightarrow (40, 35, 30)$

Give 3 characteristics to cluster 1

extra classes to cluster 2.

Outlier detection

↓
 ↳ finding data points that are very different from normal data
 Unusual / Rare
 abnormal value
 also called anomaly detection

Ex • 96 most students score 60-80 marks
 • one student score 5
 That 5 is an outlier

methods

Application fraud detection

medical diagnosis

network security

1. Statistical method
 check how far value from mean

2. IQR (Box plot method)

3. Distance based method
 very far from cluster - outlier

K-means clustering

↳ used to divide data into K number of clusters

$K \rightarrow$ no of clusters you want

means $\rightarrow$ Average / Centre of cluster + centroid

$\Rightarrow$ make clusters where points are close to centroid, separate from each other

works

1. choose K
Ex $\rightarrow$ 2 clusters
2. initialize centroids
pick random points as centres
3. assign points
each datapoint $\rightarrow$ nearest centroid
4. update centroid
Take mean of cluster points $\rightarrow$ new centroid
5. Repeat until centroids stop changing

Adv

simple fast
work well for large datasets
easy to understand

Limitations

- $\rightarrow$ Need to choose K manually
- $\rightarrow$ Sensitive to outliers
- $\rightarrow$ Work best with round shaped clusters

Applications

- Customer segmentation
- Image compression
- Market analysis

Hierarchical clustering — clusters in
tree like structure
↓
called Dendrogram

Types

Agglomerative
(Bottom → Up)
Start: Every data point = separate cluster
Then: merge closest cluster
step by step

Ex forming friends
group gradually

Divisive (Top → Down)

Start: All data = one cluster

Then: Split into smaller
clusters

Ex dividing class into
sections

work

1. Treat each point as cluster
2. Find closest two clusters
3. Merge them
4. Repeat until one cluster remains

Ex Dataset

90, 85, 40, 35

Step 1 clusters

(90) (85) (40) (35)

Step 2 merge closest

85 and 90 → merge

clusters

{85, 90} (40) (35)

Step 3 merge next closest

35 and 40 → merge

clusters

{85, 90} {35, 40}

Step 4 final merge

All → one cluster

Adv

no need to choose K initially

easy visualization

re hierarchy of clusters

disadv

→ slow for large datasets

→ memory heavy

Application

→ Biology

→ Document Clustering

→ image segmentation.

Density based Clustering

↳ group based on dense regions and separate sparse regions

- If many points are close $\rightarrow$ cluster
- If point is alone $\rightarrow$ noise/outlier

Density based Spatial Clustering of Applications with Noise) algo

$\rightarrow$ - 2 main parameter

- 1. ϵ (E)

Radius distance around a point

- 2. minpts

minimum points needed to form dense region

Types of points

$\rightarrow$ Core point
has many neighbors inside epsilon radius

$\rightarrow$ Border point
near core point but fewer neighbors

$\rightarrow$ Noise point
Far from cluster $\rightarrow$ outlier

Working

$\rightarrow$ 1 Pick random point

$\rightarrow$ 2 Count neighbors inside ϵ

$\rightarrow$ 3 If neighbors $\geq$ minpts $\rightarrow$ Start cluster

$\rightarrow$ else $\rightarrow$ mark noise

4 Expand cluster using connected core points

Ex $Eps \rightarrow 2 \text{ units}$

$\text{minpts} \geq 3$

If point has

3 + neighbors $\rightarrow$ Core

1-2 $\rightarrow$ border

0 $\rightarrow$ noise

Adv

$\rightarrow$ Find irregular shape clusters

$\rightarrow$ Automatically find no. of clusters

$\rightarrow$ Detect outliers

disadv

$\rightarrow$ Hard to choose Eps value

$\rightarrow$ Struggle if density varies

Application

$\rightarrow$ Fraud detection

$\rightarrow$ GPS location

$\rightarrow$ Anomaly detection

Association

Find interesting relations among variables within a large datasets

- Find dependency of one data item on another and map accordingly to generate max profit

Ex Market Basket analysis, Continuous production

Parametric Models

- Assume fixed functional form
- Learn finite steps of parameter
- Fast, simple, require less data
- Risk of underfitting
- Ex Linear Regression, Logistic

Non Parametric

- No fixed assumption about data distribution
- Complexity grow with data
- Flexible can model complex pattern
- Ex Decision Trees, Random Forest

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Accuracy Score Percentage of correct prediction

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total No of predictions}}$$

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

↳ how many of predictions we said are positive actually correct

$$\frac{TP}{TP + FP}$$

$$\text{Precision} = \frac{\text{Correct Yes}}{\text{All Yes predictions}}$$

Recall

↳ out of all actual Yes cases how many did the model correctly find

$$\frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{\text{Correct Yes}}{\text{All Actual Yes}}$$

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F1 → balance b/w Precision and Recall

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Silhouette score

↳ metric to evaluate clustering performance

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i); b(i)\}}$$

$a(i)$ → Average distance of point i to points in same cluster

$b(i)$ → Average distance of point i to points in nearest other cluster

Distance based method

1. Euclidean distance outlier
Point far from other outlier
2. Mahalanobis Distance
Considers correlation b/w variables